Development of ML Model for Catalyst Development in CO and CO_2 Hydrogenation to Ethanol Using Literature Data

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Application of carbon capture and utilization processes is a vital part of a circular economy.[1] Production of ethanol from the captured CO_2 is considered a key process since the ethanol is an important building block in chemical industry and sustainable aviation fuels.[2] The high global demand for such a process can be deducted from the increasing amount of research articles on the topic, which is closely correlated with oil prices (Figure 1a). These papers and respective patents report a vast number of different catalysts for CO and CO_2 hydrogenation to ethanol (Figure 1b).

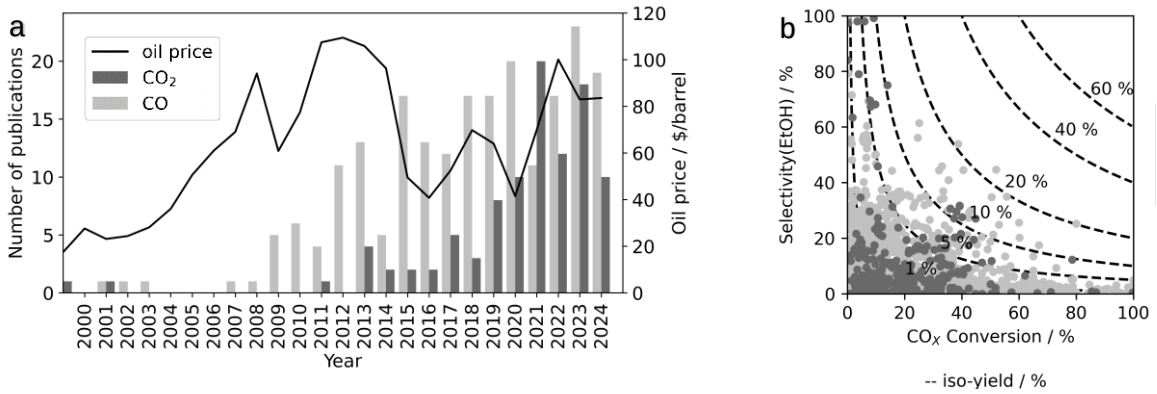

Figure 1. a) Number of publications for the conversion of CO or CO_2 to ethanol in comparison with the oil price b) Performance overview of CO_X to ethanol catalysts reported in literature.

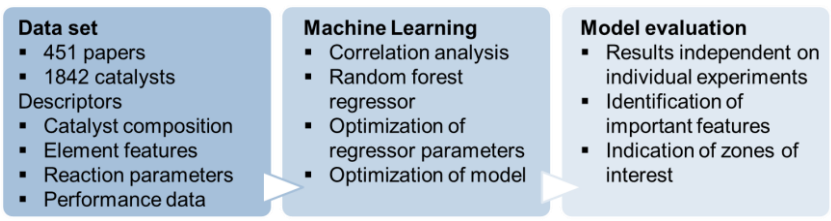

Figure 2. Schematic representation of the workflow.

While synthesizing and testing of individual catalysts is time consuming, analyzing the reported literature data with statistic methods can open new doors for enhanced catalyst development. Such investigation, known as meta-analysis, can help identifying important catalyst and condition features especially when combined with machine learning.[3-5] In the presented work, we report a meta-analysis on a dataset curated from 451 papers with 1842 different catalysts and condition variations for individual

catalysts. The collected datasets were brought into a unified structure, in which main synthesis, performance and test descriptors were identified and documented for each study (Figure 2). Furthermore, the data was enriched with elemental features. To apply a random forest regressor algorithm to the data set, different characteristics of the data needed to be taken into account. One characteristic is the heterogenicity of the dataset stemming from the broad variety of catalysts and reaction conditions making sub-datasets necessary. Additionally, the original sources often lacked the necessary metadata for a detailed description, so simplification was needed. Furthermore, the data is highly clustered due to the authors specific way of testing and reporting, which was considered for the splitting into train and test data**.**

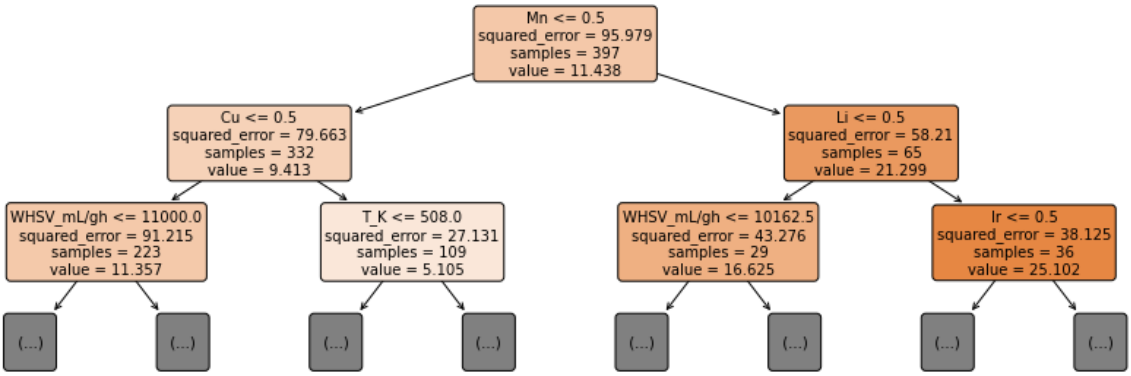

Figure 3. Decision tree for CO conversion with S(EtOH) as performance descriptor and catalyst composition and reaction conditions as descriptors.

For a data set considering CO conversion based on the reaction conditions and catalyst composition the most relevant features leading to the highest ethanol selectivity are Mn, Li and Ir corresponding to promotors well known for Rh catalysts (Figure 3).[6] On the other branch Cu is the next node explainable by its selectivity towards methanol.[7] Conducting systematic test with the regressor on different sub-sets of the data and different descriptors the prediction can be further enhanced. With those results important catalyst and condition features can be identified while being independent on individual experiments. This enables the identification of interesting feature spaces for future experiments.

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